



Deep Learning Assignment

Diploma in DS / IT

Oct 2024 Semester

ASSIGNMENT 2

(40% of DL Module)

13 Jan 2025 – 15 Feb 2025

Submission Deadline:

Presentation: 15 Feb 2025, 11:59PM

Report: 15 Feb 2025, 11:59PM

Penalty for late submission:

10% of the marks will be deducted every calendar day after the deadline.

NO submission will be accepted after **22 Feb 2025, 11:59PM.**

A. Assignment Specifications

Problem 1 (Group: 4-5 students per group) (60%)

1.1 Objective

Build a sentiment analysis model to predict the App review score based on the Google Play Store reviews.

1.2 Dataset

Each group need to collect the Google Play Store reviews by themselves.

- First, each student needs to select one google play store App and collect the review data using **google_play_scraper**
 - the sample codes are provided in “**data_collection.ipynb**”
 - the instructions on how to get the App ID is provided in Appendix

1.3 Suggested Tasks

You are recommended to tackle this problem by following the steps below. A Jupyter Notebook (**Assignment_2_p1_[student_name].ipynb**) is provided for you to write the codes based on these steps. Each student will build their own models and save their own best model and compare against the other team mates. The team will decide on the best trained model and the **Team Leader** will run the final model to make further predictions in Step 4 below.

Step 1 – Data Loading and Processing (Group)

- Data Collection
 - Each student needs to research online and select an App
 - Based on the App each student is assigned, collect the review data from Google Play Store
 - Extract the content and score columns from the review data
- Data Processing
 - Convert the content and score into numeric tensors (X & y)
 - It may be helpful to cleanse the content before encoding them into numbers
 - Any data cleaning or transformation has to be done as a group
- Data Sampling
 - Split the dataset (X & y) into training set (X_train & y_train) and testing set (X_test & y_test).

Step 2 – Develop a Sentiment Analysis Model using Training Data (Individual)

- Each student is required to develop **at least One model**, which should have:
 - Word Embedding Layer: You can use the pre-trained word embeddings (e.g. GloVe), or simply train the word embeddings using the provided dataset.

- RNN Layers (e.g. LSTM, GRU and etc.) or 1D Conv Layer or other layer suitable for sentiment analysis task
- During the training phase, please split the training data into training samples and validation samples. (**Hint:** use `validation_split` in `model fit()` function)
- You are suggested to follow the universal machine learning workflow to develop the model and improve the performance by tuning the model's hyperparameters, i.e.
 - Start with a baseline model
 - Scale up the model until it overfits
 - Regularize the model and tuning the model hyperparameters
- Analyze the Model Performance during training phase
 - **Remember to record all the model performance curves** (i.e. training and validation accuracy, training and validation loss scores) during training phase

Step 3 – Evaluate the Models using Testing Data (Individual & Group)

- Evaluate the Models Performance using `X_test` & `y_test` from other team members.
- Compare and discuss the models performance
- Recommend the best model.
- You might want to use the following table to tabulate the results.

	App 1 Name Test Result	App 2 Name Test Result	App 3 Name Test Result	App 4 Name Test Result	App 5 Name Test Result
Student 1 Name Model					
Student 2 Name Model					
Student 3 Name Model					
Student 4 Name Model					
Student 5 Name Model					

Note: for table above, key in name of Student and App

Step 4 – Use the best Model to make prediction (Group)

- Use the `input()` function to record the user input
- Convert the user input into numeric tensor
- Feed the numeric tensor into your model and see whether the model can output a correct emoticon

1.4 Report Format & Content Guidelines

Write a report with the following sections and content guidelines. You are free to include other relevant information you deem necessary in the sections.

(Note: For a page with 1-inch margins, 12-point Arial font, and minimal spacing elements, a good rule of thumb is **500 words** for a single-spaced page)

	Suggested Report Sections & Content Guidelines	Word Count
1.	Overview (<u>Group</u>) • Describe the problem, the objective, and the approach.	Min: 500 words Max: 1000 words

2.	Data Collection and Processing (<u>Group</u>) - Describe why this set of apps was selected - Describe how the data was cleansed and converted into numeric tensors - Describe how the data was sampled and why the need to sample the data	Min: 1000 words Max: 2000 words
3.	Develop the Sentiment Analysis Models (<u>Individual: One Section per Student</u>) - Describe how the models were built, trained and tuned to improve the model performance - Analyze the model performance during training phase	Per Student Min: 500 words Max: 1000 words
4.	Model Evaluation using Testing Data (<u>Individual & Group</u>) - Analyze the model performance during testing phase - Compare and discuss the models' performance - Recommend the best model and explain why	Min: 500 words Max: 1000 words
5.	Use the Best Model to Make Prediction (<u>Group</u>) - How to apply the model on a real-life text input - Explain and Analyze the model prediction	Min: 500 words Max: 1000 words
6.	Summary (<u>Group</u>) Summarize the models performance and provide suggestions for further improvement.	Min: 500 words Max: 1000 words

Problem 2 (Individual) (40%)

2.1 Objective

Implement a Recurrent Neural Network (RNN) to create an English language character generator capable of building semi-coherent English sentences from scratch, by building them up **character-by-character**. In particular, we will be using "THE ADVENTURES OF SHERLOCK HOLMES by SIR ARTHUR CONAN DOYLE".

We can train a deep learning model to generate text automatically, character-by-character, by showing the model many training examples so that it can learn a pattern between text inputs and potential character outputs. With this type of text generation, each input is a string of valid characters like this one:

"there was but one woman to him, and that woman was the la"

The corresponding output is the next character in the sentence - which is 't' (since the complete sentence is 'there was but one woman to him, and that woman was the late Irene Adler). We need to show a model many such examples in order for it to make reasonable predictions.

2.2 Dataset

Download the dataset from POLITEMall:

- **holmes.txt** : the full text of “THE ADVENTURES OF SHERLOCK HOLMES by SIR ARTHUR CONAN DOYLE”

2.3 Suggested Tasks

You are recommended to tackle this problem by following the steps below. A Jupyter Notebook (**Assignment_2_p2_[student_name].ipynb**) is provided for you to write the codes based on these steps.

Step 1 – Data Loading and Processing

- Data Loading
 - Open and read the txt file
 - Check the total number of characters in the original text
- Data Processing
 - Perform basic data cleansing by removing unnecessary characters
 - Identify a list of unique characters and punctuations in the clean text
 - Prepare data into training text and labels (X & y) using the “sliding window” method
 - Perform one-hot encoding on X & y and convert them into binary arrays

Step 2 – Develop Sequence Generator Model

- Each student is required to develop **at least ONE model**, which must consist of RNN Layers (e.g. LSTM, GRU and etc.)
- During training phase, please split the data into training samples and validation samples. (**Hint**: use `validation_split` in `model fit()` function)
- You are suggested to follow the universal machine learning workflow to develop the model and improve the model accuracy, i.e.
 - Start with a baseline model
 - Scale up the model until it overfits
 - Regularize the model accordingly
- You are also suggested to use the trained models to generate new texts (e.g. 400 characters) applying proper sampling strategy and adjust the sampling function parameters (e.g. temperature) accordingly.
- Analyze the Model Performance during training phase
 - The model performance will be a combination of accuracy and the quality of generated texts. Feel free to develop your own metrics of evaluation on the quality of the generated texts.

- **Remember to record all the model performance curves** (i.e. training and validation accuracy, training and validation loss scores) and **the generated texts** during training phase

Step 3 – Use the developed Model to generate texts

- Use the input() function to record new input from user
- Encode the input using one-hot encoding
- Feed the encoded input into your model and let the model generate 400 characters
- Justify whether the generated texts make sense to you

2.4 Report Format & Content Guidelines

Write a report with the following sections and content guidelines. You are free to include other relevant information you deem necessary in the sections.

(Note: For a page with 1-inch margins, 12-point Arial font, and minimal spacing elements, a good rule of thumb is **500 words** for a single-spaced page)

	Suggested Report Sections & Content Guidelines	Word Count
1.	Overview • Describe the problem, the objective, and the approach.	Min: 500 words Max: 1000 words
2.	Data Loading and Processing • Describe how the data was loaded and cleansed • Describe how the data was prepared and performed one-hot encoding	Min: 500 words Max: 1000 words
3.	Develop the Sequence Generator Model(s) • Describe how the models were built, trained and tuned • Analyze the model performance, including both accuracy and the quality of generated texts	Min: 1000 words Max: 2000 words
4.	Use the developed Model to Generate Texts • How to apply the model on a real-life text input • Analyze the generated texts and see whether it makes sense	Min: 500 words Max: 1000 words
5.	Summary • Summarize the model performance and provide suggestions for further improvement.	Min: 500 words Max: 1000 words

B. Recorded Presentation

Problem 1 (Group)

Each group will be allotted a total of 25 minutes to present using slides. You are strongly suggested to add in screenshots, diagrams, images into the slides and practice the presentation in advance to make sure the group can complete within 25 mins.

Problem 2 (Individual)

Each individual will be allotted a total of 5 minutes to present using slides. You are strongly suggested to add in screenshots, diagrams, images into your slides and practice the presentation in advance to make sure you can complete within 5 mins.

The recorded presentation must be through MS Teams with all student's camera on. The submission will be made by submitting the link in the comment section of the Assignment 2 submission. Penalty will be applied to non-conformance.

C. Deliverables

For this assignment, you must submit all the following:

Problem 1 (Only Team Leader needs to submit)

1. A deck of Final Presentation Slides
 - **One** final deck of slides that includes the Group and Individual Presentation Slides for Problem 1.
2. A Final Softcopy Report
 - The Final Report includes the Group Report for Problem 1.
 - Use the Cover Page and insert the link to your team Video recorded presentation
3. A zip file with Jupyter Notebook Files (.ipynb)
 - The zip file includes **Each** member's Jupyter Notebook for Problem 1.
4. Create a folder name **[Class]_[Group]** and include items 1 to 3 above into it and Zip up the folder for submission in POLITEMall.

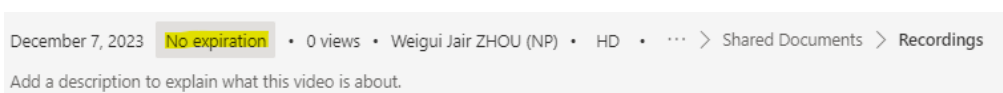
Problem 2 (Individual submission)

1. A deck of Final Presentation Slides
 - **One** deck of slides that includes the Presentation Slides for Problem 2.
2. A Final Softcopy Report
 - The Final Report includes the Report for Problem 2.
 - Use the Cover Page and insert the link to your team Video recorded presentation
3. A Jupyter Notebook File (.ipynb)
 - The file includes the Jupyter Notebook for Problem 2.

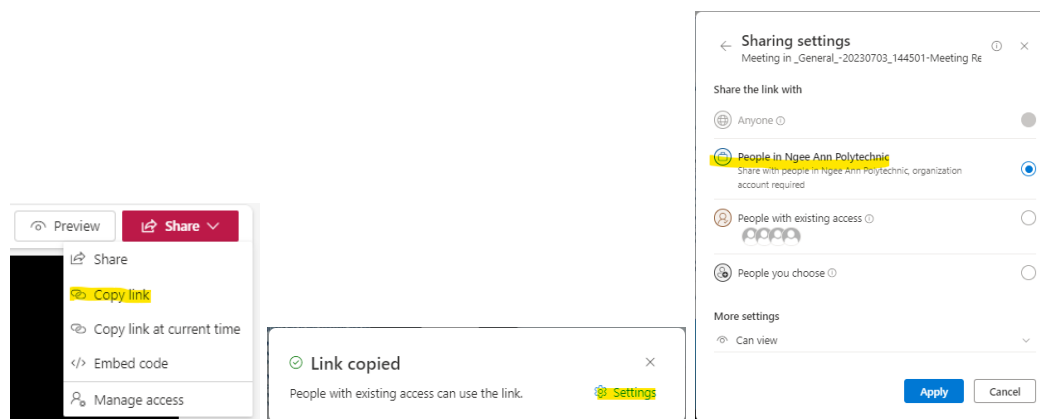
The link to your recorded presentation

- Extend your Teams recording expiration date to **No expiration**

Open your video recording in a browser and at the bottom of the screen you will see expiration date set as 30 days or lesser. Click on the date and select No expiration as shown below.



- **Edit the sharing of the recorded video** by clicking on the  at the top right corner and click on Copy link as shown below and click on  **Settings**. Ensure that you select People in Ngee Ann Polytechnic and click Apply.



- Submit the link of your recorded presentation on the **cover page (which is provided)** of your Assignment 2 Problem 1 and Problem 2 Submissions.
- **Note:** if tutors cannot access the recorded video, and students need to be reminded by tutors to reset permission, students will be penalized for non-conformance.

D. Milestone Checking

During the scheduled in-person learning (IPL) sessions in Week 17 (starting February 3, 2025) and Week 18 (starting February 10, 2025), you are required to present updates on your progress for both Problem 1 and Problem 2. This includes showcasing the details of your work and your code. Failure to do so will result in a deduction of marks.

E. Grading Criteria

A Total of 100%	Problem 1		Problem 2
	Group	Individual	Individual
Report	20%	20%	30%
Presentation	10%	10%	10%

	Grading Criteria
Presentation	a) Quality of work b) Flow of presentation based on content guidelines c) Quality of presentation slides d) Presentation and articulation skills
Final Report & Jupyter Notebooks	a) Quality of work b) Completeness of report based on suggested report sections and content guidelines

	c) Clarity of report, Quality of analysis and discussions d) Use of proper visual aids and Use of proper grammar
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F. Appendix

How to get App ID from Google Play Store?

Step 1: Go to <https://play.google.com/store/apps>

Step 2: Select the App you are interested and you can find App ID highlighted per below

